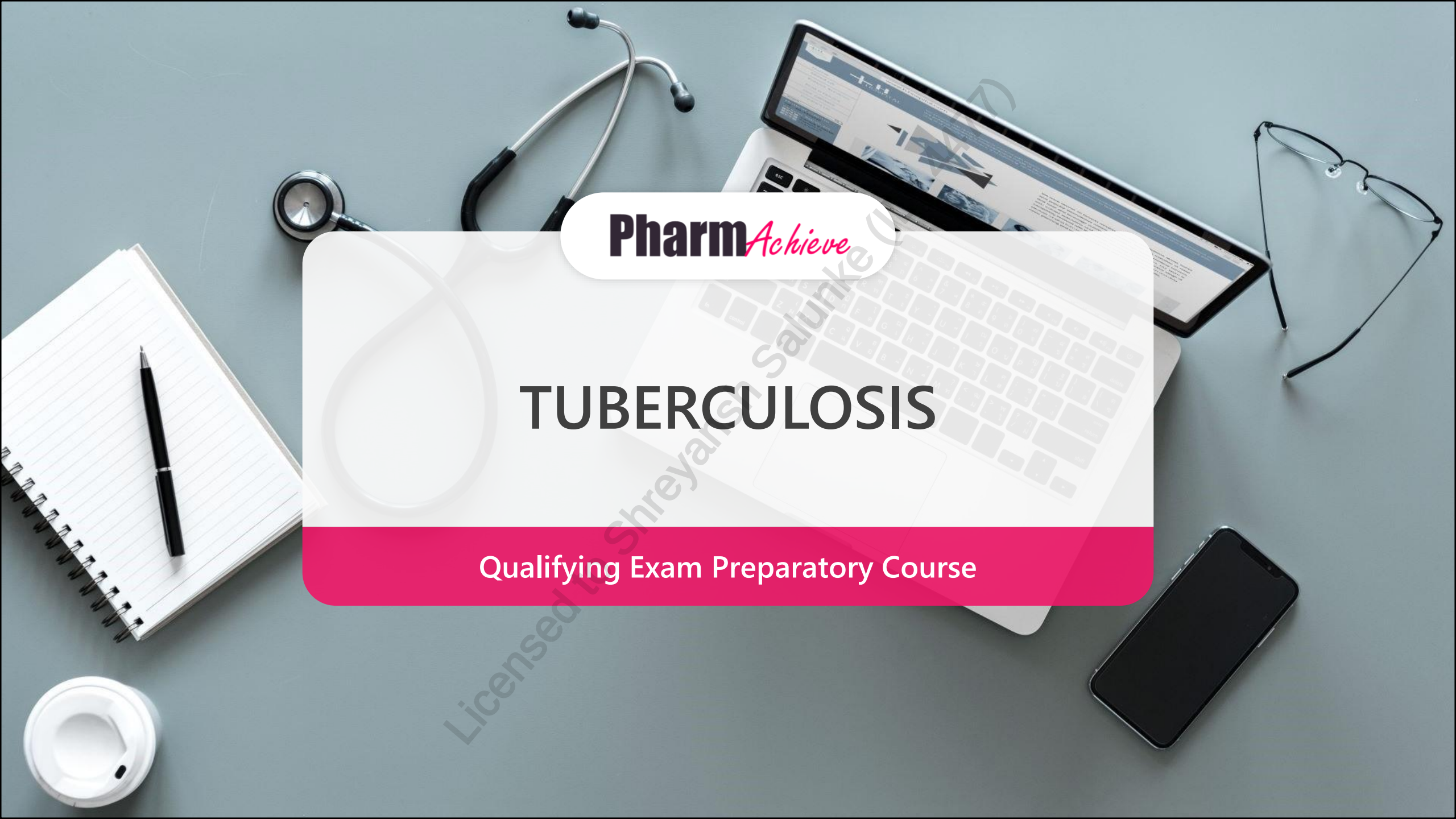


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

TUBERCULOSIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ARV	Antiretroviral
DOAC	Direct Oral Anticoagulant
DOT	Direct Observed Therapy
EMB	Ethambutol
HIV	Human Immunodeficiency Virus
INH	Isoniazid
MDR	Multidrug-Resistant
MTB	<i>Mycobacterium Tuberculosis</i>
PZA	Pyrazinamide
RIF	Rifampin
RMP	Rifampin
TB	Tuberculosis

Licensed to Shreyansh Salunke (ID: 4477)

DEFINITIONS

- Tuberculosis (TB) is a **contagious bacterial infection** that involves the **lungs** and may spread to other organs
- Infection is caused by *Mycobacterium tuberculosis (MTB)*
- Transmitted by **inhalation of aerosolized droplet nuclei**
- Transmission depends on:
 - Living conditions, immune system function, exposure, and bacterial burden
- Co-infection with HIV and TB is common and accelerates progression of both diseases
- Up to **90% do not develop any active disease** during their lifetime

EPIDEMIOLOGY

FYI

- 1×10^9 people have died of TB in past 200 years
- Estimated 10 million people developed active TB in 2019
- Top 10 causes of death worldwide
- A total of 1.4 million people died from TB in 2019 (including 208 000 people with HIV)
- Canadian national rate of approximately 4.6 cases per 100 000 people (much higher in territories, northern communities)
- Eight countries account for two thirds of the total, with India leading the count, followed by Indonesia, China, the Philippines, Pakistan, Nigeria, Bangladesh and South Africa

PATHOPHYSIOLOGY: *MYCOBACTERIUM TUBERCULOSIS* (MTB) INFECTION

- Aerobic, non-spore-forming bacillus, replicates slowly (q24h)
- Thrives in areas of high oxygen tension: lung apices, renal parenchyma, growing bones
- Cutaneous TB (1-2% cases) can occur through direct skin inoculation of MTB, endogenous or hematogenous spread
- Can also be rarely transmitted through ingestion
- **Coughing/sneezing produces "droplet nuclei" containing MTB.** Infected individual must be able to produce sputum/airborne droplets
- Inhalation of particles leads to deposition on alveolar surface
- Once implanted, *M. tuberculosis* multiplies and is ingested by pulmonary macrophages
- If macrophages kill bacilli, infection is aborted. Otherwise, macrophages containing multiples of MTB ruptures and releases more bacteria into the system in a cyclical fashion
- MTB is carried by macrophages to regional lymph nodes, spreading to the circulatory system
- **Once in blood, MTB can infect any tissue or organ**
 - Normally areas with high O₂ tension (e.g. lung apex infection causes miliary TB)
- Tissue necrosis and calcification of the infected site(s) and lymph nodes may occur

PATHOPHYSIOLOGY: *MYCOBACTERIUM TUBERCULOSIS* (MTB) INFECTION

- **Primary Disease (5%):**
 - In those unable to contain infection (often young children and immunocompromised individuals) progression of disease and complications may occur in 2 – 6 months
- **Latent TB Infection (95%):**
 - After 3-8 weeks, most immunocompetent individuals develop bacilli-specific immunity, leading lymphocytes to activate macrophages that surround the tuberculosis foci and contain the infection
 - 90% of all cases do not develop any disease, 5% may have TB reactivation

PATHOPHYSIOLOGY

- **TB Reactivation:** occurs years to decades after initial infection and presents as pulmonary or extrapulmonary disease
 - *MTB* escapes granuloma and begins to multiply extracellularly
 - Immune response, release of cytokines and lysozymes into infected foci leads to tissue damage and necrosis
 - Foci becomes unstable, liquefies (caseous necrosis), and infection spreads
- **Reinfection:** repeat exposure leads to new *MTB* infection, distinct from reactivation
 - More common in highly endemic areas, uncommon cause of active TB cases in Canada

CLINICAL PRESENTATION: PULMONARY TB (MOST COMMON)

- Weight loss, fatigue, fever, night sweats, hemoptysis, pleural pain, hoarseness, cough lasting longer than 2 weeks with/without blood and/or phlegm
- Dyspnea (dyspnea requires ventilation/perfusion mismatch, but TB kills both lung and blood vessels, so dyspnea only seen in late stage of TB)
- Patchy/nodular infiltrates in apical areas of upper lobes of lung or cavitory lesions
- Dullness to chest percussion, rales
- HIV co-infection
 - Patients are less likely to have positive skin tests, cavitory lesions or fever
 - HIV causes more rapid progression and worse outcomes in patients with TB

CLINICAL PRESENTATION: EXTRAPULMONARY TB

FYI

- **Progressive decline of organ function**, low grade fever, symptoms depend on site of involvement
- **Complications in:**
 - **Genitourinary:** sterile pyuria, dysuria
 - **Gastrointestinal:** diarrhea, colitis
 - **Lymphatic:** cervical node (most common), neck mass with spontaneous drainage, inguinal node
 - **Bone:** arthritis, pain, decreased range of motion
 - **Meningitis:** headache, confusion
 - **Miliary TB**

RISK FACTORS FOR TB EXPOSURE

Men have 2x higher rate for TB than women

Location and place of birth (e.g. sub-Saharan Africa, India, Southeast Asia, China)

Recent travel to areas of high prevalence

Crowded conditions with poor air quality and ventilation

Poor access to medical care

Close contacts, residents and employees of high-risk congregate settings: correctional facilities, homeless shelters, some healthcare facilities (e.g. nursing homes, long-term care facilities for the elderly), residential facilities for AIDS patients

RISK FACTORS FOR ACTIVE TB INFECTION AFTER EXPOSURE

Decreased immunity (e.g. use of immunosuppressants, chemotherapy)

Co-morbidities (e.g. HIV/AIDS, diabetes, hematologic/head and neck cancers, silicosis, renal disease)

Children < 2 and elderly > 65

Recent TB infection

Tuberculin skin test conversion within 2 years

Smoker or use illicit drugs

Heavy drinker

Underweight

Once infected, lifetime risk of active TB is ~10% where the highest risk within first 2 years of infection

HIV AND TB COINFECTION

- 9-16x higher risk of TB compared to HIV-uninfected individuals
 - For both primary infection and reactivation of latent TB
- HIV/TB co-infection holds negative clinical impacts for both disease states
 - Declining immunity (reduced CD4) increases risk for TB progression and complications
 - TB may accelerate progression to AIDS and death
- HIV leads to depletion of TB-fighting lymphocytes and macrophages, increasing immunosuppression → greater TB risk

DIAGNOSIS

- Review of past medical history for risk factors for exposure, development of active TB, drug resistance, previous BCG (bacillus Calmette-Guérin) vaccination
- Review history of symptoms and perform physical examination
- Laboratory investigations (e.g. chest X-ray, sputum specimens, baseline CBC, ALT, AST, bilirubin, creatinine) and special procedures as needed

DIAGNOSIS: MANTOUX SKIN TEST (OR TUBERCULIN SKIN TEST)

- Assists in the diagnosis of latent TB, also used as epidemiological tool
- Tuberculin purified protein derivative is injected intracutaneously, results in 48-72 hours
- **Positive test interpreted based on size of induration or “bump/wheel” and individual risk factors**
 - False negatives (e.g. very elderly, severe illness, immunocompromised)
 - False positives (e.g. BCG vaccination)
- Interferon-gamma release assays (IGRAs) are also used to support TB diagnosis

DRUG RELATED CAUSES OF SIGNS AND SYMPTOMS

- Immunosuppressant drugs: can cause reactivation of latent infection
- Examples: monoclonal antibodies and biologicals (e.g. TNF-alpha inhibitors, infliximab)

GOALS OF THERAPY

- ✓ Eliminate *Mycobacterium tuberculosis* (MTB) infection
- ✓ Prevent disease progression
- ✓ Prevent complications and death
- ✓ Reduce transmission with proper isolation precautions
- ✓ Prevent TB reactivation
- ✓ Prevent drug resistance by using effective combination regimen

PHARMACOLOGIC THERAPY: LATENT TB INFECTION (LTBI)

- **Asymptomatic and non-infectious** with 5-10% lifetime risk of developing active TB
- Increased risk of reactivation in high-risk groups (e.g. HIV/AIDS, immunocompromised patients, increasing age)
- Rule out active TB prior to treatment
- Consider index case resistance pattern, if known
- **First line:**
 - **Rifampin** 10 mg/kg (up to 600 mg) daily x 4 months (preferred in elderly patients)
 - OR
 - **INH** 15 mg/kg (up to 900 mg) + rifapentine WEEKLY x 3 months
 - **Rifapentine is now approved in Canada!**
- **Second Line:**
 - INH daily x 9 months (INH9)
- **Alternative regimens:**
 - INH + RMP daily x 3-4 months (level 1 evidence)
 - INH daily x 6 months (less effective)
 - INH or INH + RMP twice weekly directly observed treatment (level 2 evidence – not advised)
 - Levofloxacin or moxifloxacin daily x 9 months (if index case MDR TB R to INH and RIF)

PHARMACOLOGIC THERAPY: ACTIVE PULMONARY TUBERCULOSIS

- Combination drug therapy with emphasis on adherence
- Eliminate organisms and prevent development of resistant MTB strains
- Important to know epidemiology of resistance to existing therapies
- Therapy begins with initial intensive phase:
 - 2 months of intensive bactericidal therapy to rapidly eliminate the majority of organisms and prevent drug resistance
- Then, continuation (sterilizing) phase:
 - 4-7+ months to eradicate dormant organisms and prevent relapse
- Dose Adjustments
 - Based on susceptibility data (typically 1-2 month delay until available)
 - Based on patient factors (e.g. low BMI, renal disease)



Do not use a single agent to treat active TB or add single agent to failing regimen

PHARMACOLOGIC THERAPY: ACTIVE PULMONARY TUBERCULOSIS

Rifampin (RMP) *based* regimen is standard of care

1. Intensive phase: early bactericidal action on extracellular organisms to render patient non-infective

- INH + RMP + pyrazinamide (PZA) ± ethambutol (EMB) *daily* x 2 months
 - *EMB added if isoniazid resistance $\geq 4\%$ or unknown local susceptibilities, can D/C if full sensitivity shown on susceptibility results*
- INH + RMP + EMB *daily* x 2 months (*for patients >75 years*)

2. Continuation phase: slow killing of dormant organisms to completely eradicate infection and decrease duration of treatment

- INH + RMP *daily* or *3x weekly* x 4-7 months

*Short term treatment (2+4= 6 months) only if intensive regimen includes INH + RMP+ PZA, culture conversion at 2 months, noncavitary disease, fully sensitive bacteria

PHARMACOLOGIC THERAPY

Pyridoxine (vitamin B6)

- Given to prevent peripheral neuropathy in high-risk populations (e.g. malnourished, substance-use disorders, diabetes, HIV infection, pregnant/breastfeeding, renal dysfunction)
- Consider supplementation in all patients receiving INH

Direct Observed Therapy (DOT)

- Standard of care to improve adherence and treatment completion
- Requires clinicians/trained individuals to directly observe ingestion of medication
- Consider specifically in high-risk populations: history of treatment failure and TB relapse, drug resistance, HIV co-infection, substance abuse, mental health disorders, individuals without a fixed address



The BCG vaccine is not routinely used in Canada, only 51% effective in preventing TB

TREATMENT RELAPSE/FAILURE

FYI

- Continuous persistent positive cultures after 2 months of appropriate therapy increases risk of relapse and treatment failure
- Possible causes of treatment relapse/failure:
 - Non-adherence, especially if not employing DOT
 - Drug-resistance
 - Malabsorption
 - Lab errors
 - Transient worsening in HIV patients & overwhelming disease
- Alternate agents include: amikacin, meropenem + amoxicillin/clavulanate
- Usually use at least 5 agents in MDR TB for 12+ months

REFER TO TB EXPERT
if treatment interruption,
failure/relapse, and
multiple drug resistance

SPECIAL POPULATIONS: PEDIATRIC AND EXTRAPULMONARY TB

FYI

Pediatric TB (children <5 years old with known or suspected TB)

- Initiate treatment ASAP due to increased risk of severe disease (military/meningeal TB)
- Similar regimen as adult treatments

Extrapulmonary TB

- Similar treatment to pulmonary TB
- Extended duration of treatment required for bone, joint, military, and/or CNS TB
- Routine adjunctive therapy with corticosteroids in meningeal TB: decreases mortality
- Consider high dose rifampin in CNS disease (up to 35mg/kg/d)

SPECIAL POPULATIONS: HIV CO-INFECTION

- Same principles of therapy as TB infection alone
- DOT daily therapy recommended to decrease risk of resistance and relapse
- At least 8 months of therapy required if patient not on antiretrovirals (ARVs)
- Rifamycins (e.g. RMP) decreases [C] of many ARVs – protease inhibitors (PI) and non-nucleoside reverse transcriptase inhibitors (NNRTIs)
 - Consider using rifabutin (least potent inducer) instead of RMP
 - Monitor drug therapy and make dose adjustments as needed
- Immediate (i.e. < 2 weeks after TB therapy started) initiation of ART suggested (UNLESS CNS TB, where delayed ART > 2 weeks should be used)
- Important to continue ARV therapy during TB treatment
- Monitor for IRIS (immune reconstitution inflammatory syndrome)
 - Symptoms include fever, malaise, and local organ reactions

SPECIAL POPULATIONS: PREGNANCY AND LACTATION

Active TB

- Do not withhold therapy as the risk of untreated TB outweighs risks to newborn
- INH + RIF + EMB → no teratogenicity
- Pyrazinamide → less data, unlikely to be teratogenic, controversy in use
- Avoid aminoglycosides and REFER to TB expert in drug-resistant cases

Latent TB

- Avoid pregnancy if receiving LTBI treatment (if possible)
- Case by case assessment if pregnant
 - If low risk: withhold treatment until 3 months postpartum to reduce risk of hepatotoxicity
 - Rifampin 4 month regimen preferred
 - If rifampin resistant, consider INH with monitoring for hepatotoxicity
 - Pyridoxine supplementation recommended

Breastfeeding is safe with INH, RMP, PZA, EMB and pyridoxine

DRUG TOXICITY IN TB TREATMENT

GI upset	Rash	Fever	Hepatitis
• Common in first few weeks • Take with food • Consider anti-emetics and acid-reducing agents	• Petechial, mucocutaneous generalized • Discontinue drug if SJS/TENS	• TB fever may persist for 2 months • Flu-like illness (drug fever), will resolve with discontinuation	• AST >3-5 times the normal limit • Stop all potential drug causes • Initiate LESS hepatotoxic TB drug regimen • Drug Risk: PZA>INH>RIF>EMB • Moxifloxacin, linezolid, amikacin, clofazimine, etc other options

SIDE EFFECTS OF TB TREATMENT

1 st line drug	Adverse effects
Isoniazid (INH)	• N/V, GI upset, drug-induced lupus, skin rash, mood changes, drug fever, encephalopathy, drowsiness, gynecomastia, pancreatitis, lymphadenopathy • Hematologic side effects, peripheral neuropathy (dose-related) • Hepatotoxicity: monitor carefully and educate patients on symptoms (ranges from asymptomatic liver enzyme elevations to hepatitis)
Rifamycins (Rifampin and rifabutin)	• GI upset, flu-like symptoms, metallic taste, myalgia, hepatotoxicity, neutropenia, leukopenia, discolouration of bodily fluids, rash (usually minor – petechial rash may suggest thrombocytopenia) • RMP specifically: confusion, ataxia, visual toxicity, acute interstitial nephritis • Major P450 enzyme inducer, monitor therapy and dose adjust • Rifabutin is a less potent inducer than RMP (use in patients on PIs/NNRTIs)
Pyrazinamide (PZA)	• Hepatotoxicity (rare if only using for 2 months) • GI upset, polyarthralgia (common), hyperuricemia (usually asymptomatic), drug fever • Hematologic (e.g. thrombocytopenia, leukopenia)
Ethambutol (EMB)	• GI upset, headache, dizziness, confusion, hallucinations, rash (SJS/TENS), hematologic, optic neuritis (monitor visual acuity at baseline and monthly), red/green vision change due to retrobulbar neuritis (dose related but rare → immediate notification/discontinuation), separate dosing from aluminum hydroxide by 4h

SELECT COMMON/RELEVANT RIFAMPIN INTERACTIONS

Drug class	Interaction
Calcium Channel Blockers (CCB)	• Reduced concentrations of CCB. May require increase in dose.
Statins	• Reduced statin concentration (atorvastatin, simvastatin, lovastatin) • Initial increase statin concentrations, followed by decrease (rosuvastatin)
Azole antifungals	• Reduced azole concentrations
Opioids	• Methadone, fentanyl concentration reductions • Potential for rifampin withdrawal included toxicity
Levothyroxine	• Reduced levothyroxine half life • Potential need to increase dose
Calcineurin inhibitors	• Reduced concentrations of tacrolimus, cyclosporine
Anticoagulants	• Decreased concentrations of warfarin, DOAC. DOAC contraindicated.
Anticonvulsants	• May decreases phenytoin, carbamazepine, phenobarbital levels, valproic acid levels
Oral contraceptives	• Will decrease systemic progesterone/estrogen concentrations, increase failure risk. Contraindicated.

MONITORING PARAMETERS

Clinical monitoring

- Monitor regularly for changes in symptoms
- Chest X-ray at diagnosis, 2 months (active TB) and at end of treatment

Smears and cultures

- Positive culture at 2 months is a predictor of relapse

Routine lab monitoring

- Complete baseline weight, bilirubin, AST/ALT, SCr, visual acuity, colour vision, Hep B/C serology, HIV and HbA1c
- Monthly: CBC, AST/ALT, SCr and bilirubin monthly
- Visual acuity monitoring and contrast assessment monthly if ethambutol is used

TAKEAWAYS

- Tuberculosis causes significant morbidity and mortality. Characterizing latent versus active disease is important for public health and treatment
- **Latent TB infection:** 1st line – rifampin daily x 4 months **or** isoniazid + rifapentine weekly x 3 months
- **Active TB infection:**
 - Intensive phase: INH + RMP + pyrazinamide + ethambutol *daily* x 2 months
 - Continuation phase: INH + RMP daily or 3x/week x 4-7 months.
- Prevent peripheral neuropathy with pyridoxine (Vit B6) supplement in high-risk patients on INH
- **DOT** is recommended if concerns of non-adherence; especially if treatment failure, disease relapse, drug resistance organism, comorbid conditions, HIV coinfection
- GI upset (N/V, abdominal pain) can be minimized by educating the patient to take medications with food. Assess for drug-induced hepatitis
- Assess **LFTs** at baseline, monthly and/or when symptomatic. Screen for drug interactions!
- Ensure medication adherence with patient-centered approach

CASE 1

ST is a 77-year-old male presenting to the hospital with blood-tinged sputum. ST noted a three-month history of night sweats, coughing, weight loss (5 kg, unintentional) and fatigue. He has no sick contacts and no significant travel history over the last 30 years. He immigrated from India in 1978. While in the hospital, ST remains hemodynamically stable without hypoxia. He undergoes an induced sputum sample, which finds AFB (3+), and PCR confirms the presence of *M. tuberculosis*. His chest X-ray reveals no cavitary lesions, but some scattered granulomas are present. He is a retired electrician who lives alone; he does not drink alcohol or smoke. He has a past medical history of dyslipidemia, non-alcoholic steatohepatitis and hypertension. He takes atorvastatin 40 mg PO daily, ibuprofen 400 mg PO Q6H PRN, indapamide 5 mg PO QAM, saw palmetto daily, and ramipril 10 mg PO daily. He has no allergies to any medications.

1. Which of the following baseline assessments is the most important prior to initiating therapy?
 - a) ALP
 - b) Visual acuity
 - c) ECG
 - d) ALT



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2. What is the best initial choice for ST's anti-TB regimen?
- a) Rifampin+Isoniazid+Ethambutol+Pyrazinamide
 - b) Rifabutin+Isoniazid+Ethambutol+Moxifloxacin
 - c) Rifampin+Isoniazid+Pyrazinamide
 - d) Rifampin+Isoniazid+Ethambutol



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3. Assuming the MTB isolate is susceptible to the regimen chosen and that his repeat 2-month sputum culture is negative, what is the best choice for total duration of therapy for ST?
- a) 4 months
 - b) 6 months
 - c) 7 months
 - d) 9 months



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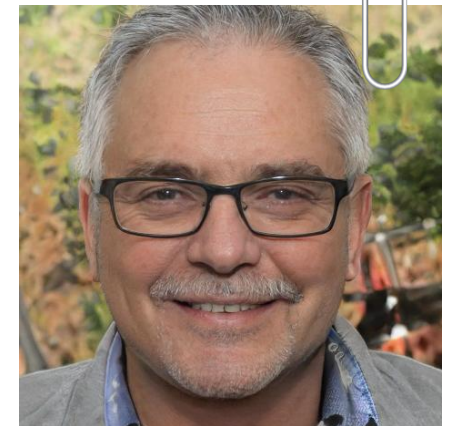
4. What monitoring would be suggested for ST prior to hospital discharge?
- a) Visual acuity and colour vision monthly for duration of therapy
 - b) Sputum for TB smear monthly
 - c) SCr monthly
 - d) Hepatitis B serology monthly



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5. Which of ST's home medications interact with his consolidation regimen and what is the appropriate management strategy?
- a) Isoniazid interacts with ramipril. Change indapamide to candesartan
 - b) Rifampin interacts with indapamide. Increase indapamide dose to 10mg daily
 - c) Rifampin interacts with atorvastatin. Change rifampin to rifabutin
 - d) Rifampin interacts with atorvastatin. Change atorvastatin to rosuvastatin



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CHANGE LOG

March 2025

- Content reviewed, minor grammatical edits made
- Slide 2: Added 'covered in this presentation' slide
- Slide 3: Added legend

July 2025

- Deleted objectives slide
- Added 'FYI' to some slides
- Deleted military TB slide
- Slide 27 emphasized hepatitis
- Amended case

Aug 2025

- Slide 23: Removed agents no longer marketed on DPD

October 2025

- Slide 19: Added that rifapentine is not approved in Canada (it can still be accessed in special circumstances)
- Slide 26: added PZA and pyridoxine to breastfeeding
- Slide 37: Updated references

February 2026

- Slide 2: Updated NAPRA competencies
- Combined slides 6 and 7 into one slide (slide 6)
- Slide 18: Updated that rifapentine got approved in Canada (in Feb 2026)
- Slide 30: Updated takeaway slide